

Clarkson Institute Summer School: Examples Sheet 5 - Conjugacy and normality

Please send all comments and corrections to jmm232@cam.ac.uk.

Conjugate elements and normal subgroups

1. Starting from first principles, partition the elements of D_6 into conjugacy classes, explaining your reasoning both algebraically and geometrically. Hence, identify the normal subgroups of D_6 .
2. Partition the elements of Q_8 into conjugacy classes. Hence, identify the normal subgroups of Q_8 .
3. Show that if $H \leq G$ is a subgroup of a group G which only has two distinct left cosets in G , then H must be normal.
4. The *centre* $Z(G)$ of a group G is the set of all elements which lie in singleton conjugacy classes,

$$Z(G) = \{z \in G : gzg^{-1} = z \text{ for all } g \in G\}.$$

- (a) Show that $Z(G)$ is an Abelian normal subgroup of G .
- (b) Show that if H is a normal subgroup of G with $|H| = 2$, then $H \leq Z(G)$.
- (c) Determine $Z(D_4)$.

Quotient groups

5. Using your results from Questions 1 and 2, construct all possible quotients of D_6 and Q_8 . Identify standard cyclic and dihedral groups to which the quotients are isomorphic.
6. Let m be a reflection in D_4 . Show that $\{e, m\}$ is not a normal subgroup. Construct its left cosets and right cosets, and hence show that coset multiplication is not well-defined in this instance.
7. In theoretical physics, a *Lorentz boost* is a symmetry transformation of $(1+1)$ -dimensional spacetime represented by the matrix:

$$\begin{pmatrix} \cosh(\phi) & -\sinh(\phi) \\ -\sinh(\phi) & \cosh(\phi) \end{pmatrix},$$

where ϕ is a real parameter, called the *rapidity* of the Lorentz boost. It represents a transformation between a stationary frame of reference, with coordinates (x, ct) , and a moving frame of reference, (x', ct') .

- (a) Show that the set H of Lorentz boosts forms a group.
- (b) The set of Lorentz boosts is now extended with time-reversal symmetry T , and parity inversion P , represented by the matrices:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad P = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Let G be the group generated by the elements of H , together with T and P . Show that $H \leq G$ is a normal subgroup of G . Is the group K , generated by T and P only, a normal subgroup of G ?

- (c) Construct the quotient group G/H , and identify it with a standard group.

8. By naturally identifying C_n with a subgroup of C_{nm} , prove that:

$$\frac{C_{nm}}{C_n} \cong C_m.$$

Deduce the structure of all possible quotients of C_{24} .

9. Show that both H and K may be naturally identified with normal subgroups of $H \times K$. Hence, prove that:

$$\frac{H \times K}{H} \cong K \quad \text{and} \quad \frac{H \times K}{K} \cong H.$$

Common misconceptions about quotients

10. Is it true that for any group G with a normal subgroup $H \trianglelefteq G$, we must have:

$$\frac{G}{H} \times H \cong G?$$

Give a proof or counterexample as appropriate.

11. If $K \trianglelefteq H$, and $H \trianglelefteq G$, is it true that $K \trianglelefteq G$? Provide a proof or counterexample as appropriate.
12. Let $H, K \trianglelefteq G$ be normal subgroups of a group G with $H \cong K$. Is it true that $G/H \cong G/K$? Provide a proof or counterexample as appropriate.